CS166 Final Project Report

Modeling the male loneliness epidemic via social networks and algorithmic exposure

Executive summary

I built an agent based Monte Carlo simulation to model a feedback loop between social isolation, algorithmic recommendation of red pill content, and ideological drift among young men. Each person is a node in a friendship network with weighted ties that represent emotional support. Isolation increases the chance that platforms recommend red pill content, and red pill exposure increases ideological drift. Friends can pull drift back toward healthier views, but friendships weaken when people's opinions diverge and when isolation rises. This makes social engagement fall and isolation rise further.

I compare three interventions against a baseline. The first intervention reduces algorithmic amplification of red pill content for isolated users. The second intervention increases δ, so that social engagement reduces isolation more strongly. The third intervention increases β_{friend}, so that friends have a stronger corrective influence. In this run, mean drift often saturates near its upper bound by the end of the horizon because the model clips drift to $[0, 1]$, so isolation and red pill exposure are especially informative for comparing strategies. The largest improvement comes from increasing δ. This does not mean that support appears from nowhere. In the model, δ controls how effectively existing social engagement reduces isolation, and social engagement depends on both friendship strength and current isolation. The results therefore support a combined recommendation. Communities should invest in sustained support that creates meaningful relationships, and they should also reduce the barriers that prevent isolated people from using support. Algorithm changes that reduce amplification for highly isolated users complement this approach.

Model

Each agent has four dynamic variables.

Isolation: $I_i(t) \in [0, 1]$.

Red pill exposure: $R_i(t) \in [0, 1]$.

Ideological drift: $D_i(t) \in [0, 1]$.

Social engagement: $S_i(t) \in [0, 1]$.

Friendships are modeled as weighted edges $w_{ij}(t)$ that represent emotional support. I only model normal friendships, not red-pill communities since most men affected by red-pill content aren't the ones on the podcasts profiting off of this loneliness epidemic.

The main experiments use an Erdos Renyi random graph with $n = 800$ and an expected mean degree close to twelve. Each edge weight is initialized uniformly between $w_{\min}$ and $w_{\max}$. Initial isolation, initial red pill exposure, and initial drift are drawn from Beta distributions.

Red pill exposure update rule

$$R_i(t+1) = (1-\alpha)R_i(t) + \alpha E_i(t), \qquad E_i(t) \sim \text{Bernoulli}(p_{alg}(I_i(t))),$$

$$p_{alg}(I_i(t)) = p_0 + (p_{\max} - p_0)I_i(t)$$

where

- $R_i(t)$ is red pill exposure at time t.
- $R_i(t+1)$ is red pill exposure at time $t+1$.
- $\alpha \in [0,1]$ controls how much new exposure updates R_i.
- $E_i(t)$ is the exposure event at time t. $E_i(t)$ takes values in $\{0,1\}$.
- $p_{alg}(I_i(t))$ is the algorithmic recommendation probability for red pill content. $p_{alg}(I_i(t)) \in [0,1]$.
- $I_i(t)$ is isolation at time t.
- $p_0 \in [0,1]$ is the baseline recommendation probability.
- $p_{\max} \in [0,1]$ is the recommendation probability for a fully isolated agent.

This rule captures how platforms steer more isolated men toward extreme content. When isolation rises, the algorithmic recommendation probability for red pill content rises, and red pill exposure increases. The parameter α controls how fast exposure responds to new content. The function $p_{alg}(I)$ makes the algorithm respond to isolation, which reinforces the spiral.

Ideological drift update rule

$$D_i(t+1) = \text{clip}_{[0,1]}\Big(D_i(t) + \beta R_i(t)\big(1 - S_i(t)\big) - \beta_{friend}\frac{1}{k_i(t)}\sum_{j \in N(i,t)} w_{ij}(t)\big(D_i(t)$$

where

- $D_i(t)$ is ideological drift at time t.
- $D_i(t+1)$ is ideological drift at time $t+1$.
- $\beta > 0$ controls how strongly red pill exposure increases drift.
- $R_i(t)$ is red pill exposure at time t.
- $S_i(t)$ is social engagement at time t.
- $\beta_{friend} > 0$ controls how strongly friends counteract drift.
- $k_i(t)$ is the number of friends of agent i at time t.

- $N(i,t)$ is the set of friends of agent i at time t.
- $w_{ij}(t)$ is the emotional support weight on edge (i,j) at time t.
- $D_j(t)$ is the drift of friend j at time t.

Ideology can move toward red pill views based on exposure. This effect is stronger when social engagement is low, meaning men without regular supportive contact are more vulnerable. Ideology can also move back toward healthier views through corrective influence from friends. When friendships are strong, the weighted average of friend ideologies gently pulls drift downward. This rule formalizes the idea that drifting men can still be helped until isolation and divergence break the social tie.

Social engagement definition and update

In the implementation, social engagement is not a fixed parameter. It is recomputed each timestep from the current friendship network and the current isolation level. It rises when an agent has stronger supportive ties, and it falls when ties weaken or the agent becomes more isolated.

$$\bar{w}_i(t) = \frac{1}{k_i(t)} \sum_{j \in N(i,t)} w_{ij}(t), \qquad S_i(t) = \mathrm{clip}_{[0,1]}\Big(\bar{w}_i(t)\,(1 - I_i(t))\Big)$$

where

- $S_i(t)$ is social engagement at time t.
- $\bar{w}_i(t)$ is the mean emotional support weight among agent i's friends at time t.
- $k_i(t)$ is the number of friends of agent i at time t.
- $N(i,t)$ is the set of friends of agent i at time t.
- $w_{ij}(t)$ is the emotional support weight on edge (i,j) at time t.
- $I_i(t)$ is isolation at time t.

Isolation update rule

$$I_i(t+1) = \mathrm{clip}_{[0,1]}\Big(I_i(t) + \gamma \frac{1}{k_i(t)} \sum_{j \in N(i,t)} |D_i(t) - D_j(t)| - \delta S_i(t)\Big)$$

where

- $I_i(t)$ is isolation at time t.
- $I_i(t+1)$ is isolation at time $t+1$.
- $\gamma > 0$ controls how strongly ideological mismatch increases isolation.
- $k_i(t)$ is the number of friends of agent i at time t.
- $N(i,t)$ is the set of friends of agent i at time t.
- $\delta > 0$ controls how strongly social engagement reduces isolation.
- $S_i(t)$ is social engagement at time t.

Isolation grows when a man's worldview drifts away from his friends' worldview. Even if friendships remain intact, ideological mismatch produces emotional distance. Meanwhile, supportive social engagement reduces isolation. This captures the core

dynamic of the model. The more a man drifts, the lonelier he feels, and the lonelier he feels, the more he drifts.

Friendship weakening update rule

$$w_{ij}(t+1) = \text{clip}_{[0,1]}\Big(w_{ij}(t)\,[1 - \mu_1|D_i(t) - D_j(t)| - \mu_2\tfrac{I_i(t)+I_j(t)}{2}]\Big)$$

where

- $w_{ij}(t)$ is the emotional support weight on edge (i,j) at time t.
- $w_{ij}(t+1)$ is the emotional support weight on edge (i,j) at time $t+1$.
- $\mu_1 > 0$ controls erosion from ideological distance.
- $\mu_2 > 0$ controls erosion from isolation.
- $I_i(t)$ and $I_j(t)$ are the isolation levels of the two friends at time t.

Two forces weaken friendships over time. Ideological divergence erodes alignment and trust and makes the relationship feel strained even if it continues. Increasing isolation contributes to withdrawal and makes the relationship feel one sided. Together, these mechanisms capture how supportive ties dissolve as a man drifts further into loneliness.

Friendship cutoff condition

If $|D_i(t) - D_j(t)| > \tau$, the tie is removed with probability λ.

- $\tau \in (0,1)$ is the ideological cutoff threshold.
- $\lambda \in [0,1]$ is the probability that an above threshold tie is removed.

This implements the idea that relationships can abruptly end once divergence exceeds a threshold, but that breaking is not deterministic.

Assumptions and limitations

The model treats agents as behaviorally homogeneous, aside from differences created by network position and random initial state. This means the model does not capture heterogeneity such as age, background, or baseline mental health. This assumption can be reasonable for exploring mechanisms, but it may fail when demographic or contextual differences are the main drivers of outcomes.

The platform algorithm is simplified to a linear relationship between isolation and the probability of being recommended red pill content. Real recommendation systems depend on watch history, interactions, and platform changes over time. The simplified mechanism is useful because it makes the feedback loop explicit, but it should not be interpreted as a literal model of any specific platform.

Social engagement is represented by a proxy computed from friendship weights and isolation. In reality, engagement is influenced by time constraints, geography, and

many unobserved factors. The proxy is intended to reflect the idea that support requires both strong ties and willingness to participate socially.

The model clips state variables to the interval from zero to one. This ensures numerical stability and interpretability, but it introduces saturation effects. In particular, drift can cap out near one, which affects theory comparisons and can make final drift less informative than earlier time outcomes.

The baseline network is an Erdos Renyi random graph. This is analytically convenient for theoretical checks, but it is not fully realistic. I include an extension using a stochastic block model to introduce clustering and community structure.

Parameters and interventions

The baseline network is an Erdos Renyi graph with eight hundred nodes and expected mean degree near twelve. The parameter p_0 sets a baseline probability of red pill recommendation, and the parameter $p_{\max}$ sets the recommendation probability for fully isolated agents. The parameter α controls how quickly red pill exposure responds to new recommended content. The parameter β controls how strongly red pill exposure increases drift, and the parameter β_{friend} controls how strongly friends pull drift back toward the local neighborhood. The parameters γ and δ control how mismatch and social engagement affect isolation. The parameters μ_1 and μ_2 control how quickly friendship weights erode. The parameters τ and λ control tie breaking when divergence becomes large.

I compare three interventions against the baseline. The lower algorithm intervention reduces $p_{\max}$, which weakens algorithmic amplification for isolated users. The boost support intervention increases δ, which makes social engagement more effective at reducing isolation. The stronger friends intervention increases β_{friend}, which increases the corrective influence from friends. All interventions are evaluated by running many Monte Carlo simulations and comparing outcome distributions.

Implementation

The simulation is implemented in `CS166_final_codebook.ipynb` using object oriented design. The `LonelinessModel` class encapsulates the friendship network, the agent state variables, and the update rules. The `ExperimentRunner` class runs Monte Carlo experiments across scenarios.

I included sanity checks that ensure all state variables remain within bounds, and visualizations that show the simulation behaving as expected. The figure below shows a single baseline run and the evolution of mean isolation, mean red pill exposure, mean drift, mean social engagement, and the number of edges.

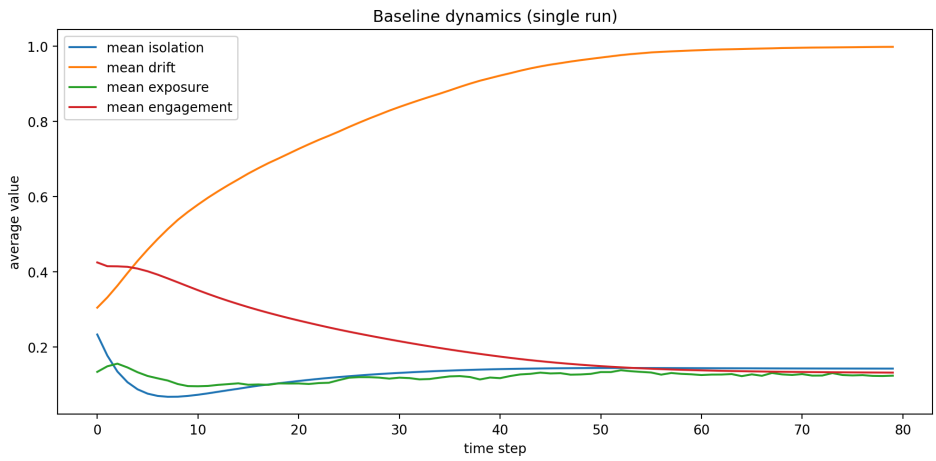

Figure 1. A single baseline run on an Erdos Renyi network. The plot shows time series of population averages and network size.

Empirical results and strategy comparison

I compare four scenarios. The baseline scenario uses the default parameterization of the model, and the other three scenarios implement interventions by changing one baseline parameter at a time. I run many independent simulations for each scenario, which yields a distribution of outcomes for each metric.

In the baseline scenario, I use an Erdos Renyi friendship network with $n = 800$ and expected mean degree close to twelve, and I simulate $T = 80$ timesteps. I chose an Erdos Renyi network because it makes the theoretical comparison tractable, and I chose the population size and degree so that averages are stable while the network remains sparse.

I initialize friendship weights uniformly between $w_{\min} = 0.3$ and $w_{\max} = 0.9$. I initialize isolation and drift using Beta distributions with parameters 2 and 5, which concentrates values below one half. I initialize red pill exposure using a Beta distribution with parameters 1 and 8, which makes early exposure uncommon and ensures that increases in exposure are driven by the algorithmic feedback loop.

I set $p_0 = 0.05$, $p_{\max} = 0.60$, and $\alpha = 0.30$ to create a baseline in which everyone has some chance of encountering red pill content, while isolated agents are much more likely to receive it and recent exposure matters more than distant exposure. I set $\beta = 0.35$ and $\beta_{friend} = 0.60$ so that red pill exposure can meaningfully increase drift while friend influence can counteract drift when ties remain strong. I set $\gamma = 0.25$ and $\delta = 0.25$ so that ideological mismatch increases isolation and social engagement reduces it at comparable scale. I set $\mu_1 = 0.25$, $\mu_2 = 0.20$, $\tau = 0.60$, and $\lambda = 0.10$ so that friendships weaken gradually and sometimes break when divergence becomes large.

The lower algorithm intervention reduces $p_{\max}$ from 0.60 to 0.35, which weakens algorithmic amplification for highly isolated agents. The boost support intervention increases δ from 0.25 to 0.40, which makes social engagement more effective at

reducing isolation. The stronger friends intervention increases β_{friend} from 0.60 to 0.85, which strengthens the corrective influence from friends.

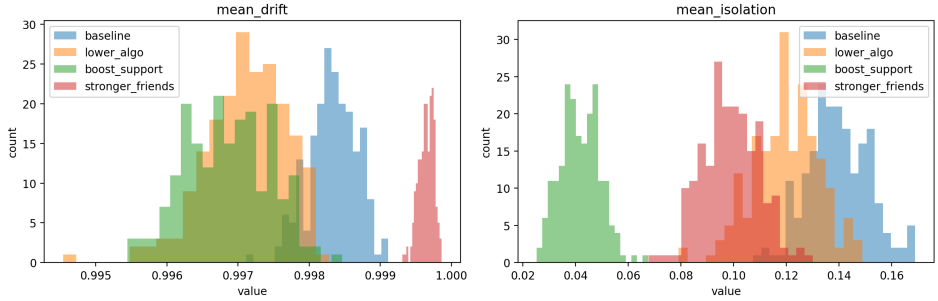

Figure 2. Monte Carlo outcome distributions on an Erdos Renyi network. The histograms show the distribution of final mean drift and final mean isolation across scenarios.

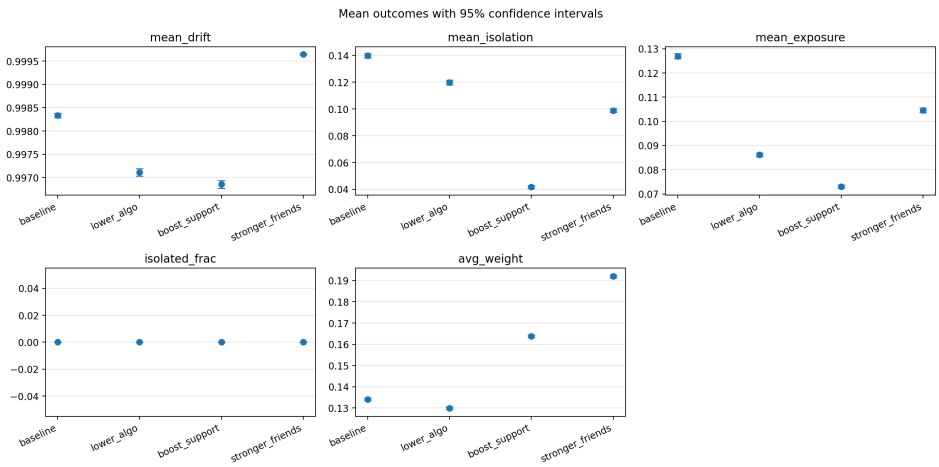

Figure 3. Expected outcomes with ninety five percent confidence intervals over Monte Carlo runs. The intervals quantify uncertainty in the estimated expected values.

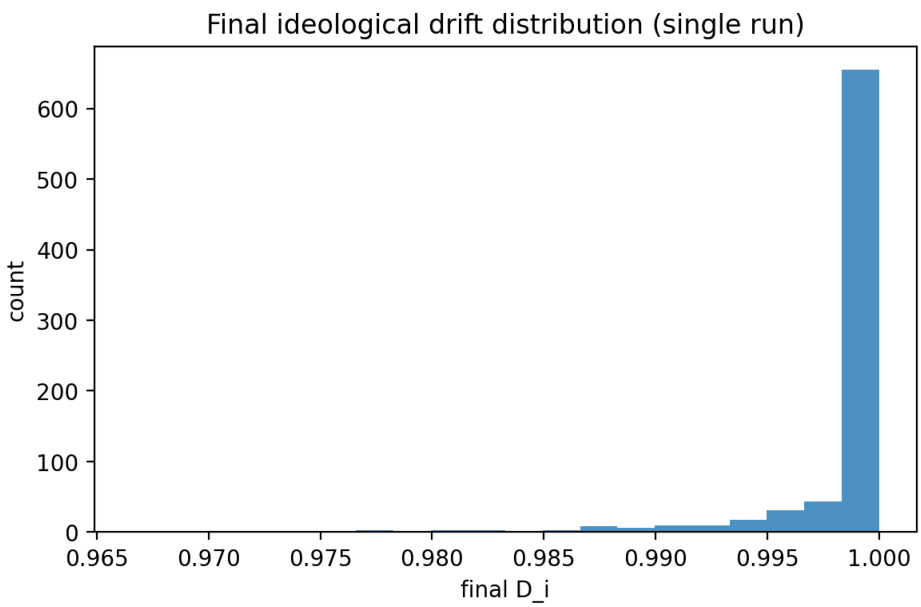

Figure 4. A zoomed view of the distribution of final mean drift. The concentration near one reflects drift saturation caused by clipping and the chosen time horizon.

In the baseline scenario, mean isolation at $t = T$ is about 0.14 and mean red pill exposure at $t = T$ is about 0.13. Mean drift at $t = T$ is typically near one because drift is clipped to $[0, 1]$ and the exposure driven term accumulates over time. Boosting support produces the largest reduction in isolation and it also lowers red pill exposure. Lowering algorithmic amplification reduces red pill exposure directly and it also reduces isolation compared to baseline. Strengthening friend influence increases average tie strength, but it does not reduce final drift under this parameterization, which suggests that reducing isolation and reducing algorithmic amplification are more direct levers in this model.

Theoretical analysis and comparison

The simulation is built on local interactions on a network, but the Erdos Renyi structure makes it possible to derive simple mean field benchmarks. These benchmarks are not exact. Their purpose is to provide a sanity check and to explain why certain patterns appear in the empirical results.

Consider the red pill exposure update. The event $E_i(t)$ is Bernoulli with mean $p_{alg}(I_i(t))$, so its conditional expectation is $\mathbb{E}[E_i(t) \mid I_i(t)] = p_{alg}(I_i(t))$. Taking expectations in the exposure recursion yields

$$\mathbb{E}[R_i(t+1) \mid I_i(t)] = (1-\alpha)R_i(t) + \alpha p_{alg}(I_i(t)).$$

A mean field approximation replaces $I_i(t)$ by the population mean $\bar{I}(t)$ and replaces $R_i(t)$ by the population mean $\bar{R}(t)$. This produces a one dimensional recursion for mean red pill exposure

$$\bar{R}(t+1) \approx (1-\alpha)\bar{R}(t) + \alpha p_{alg}(\bar{I}(t)).$$

If $\bar{I}(t)$ were approximately constant over a window of time, then $\bar{R}(t)$ would approach $p_{alg}(\bar{I})$ exponentially fast. The closed form solution of the linear recursion is

$$\bar{R}(t) \approx (1-\alpha)^t \bar{R}(0) + \big(1 - (1-\alpha)^t\big) p_{alg}(\bar{I}).$$

For drift, a rough benchmark focuses on the exposure driven term. Under exchangeability on a large random graph, the friend pull term tends to average out in expectation, so it mainly reduces variance rather than shifting the mean. This leads to the approximation

$$\bar{D}(t+1) \approx \mathrm{clip}_{[0,1]}\big(\bar{D}(t) + \beta \bar{R}(t)\,(1 - \bar{S}(t))\big).$$

The clipping is critical. An unclipped benchmark would keep increasing mean drift, but the implemented model clips each agent's drift to the interval from zero to one. Once many agents approach one, the empirical mean drift flattens. This is a major reason why a naive theory line can overshoot empirical drift.

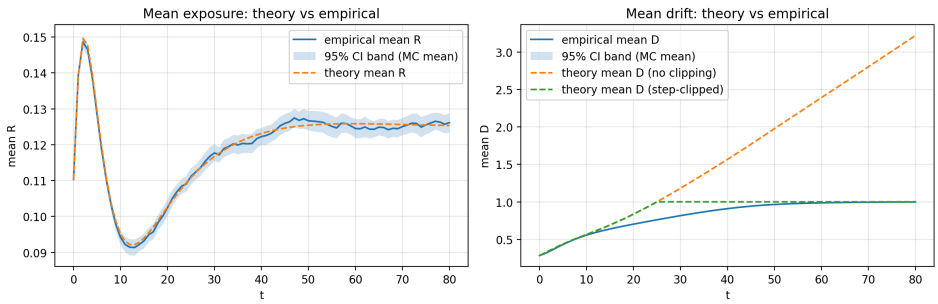

Figure 5. Empirical mean trajectories compared against a simple mean field benchmark for mean red pill exposure and mean drift.

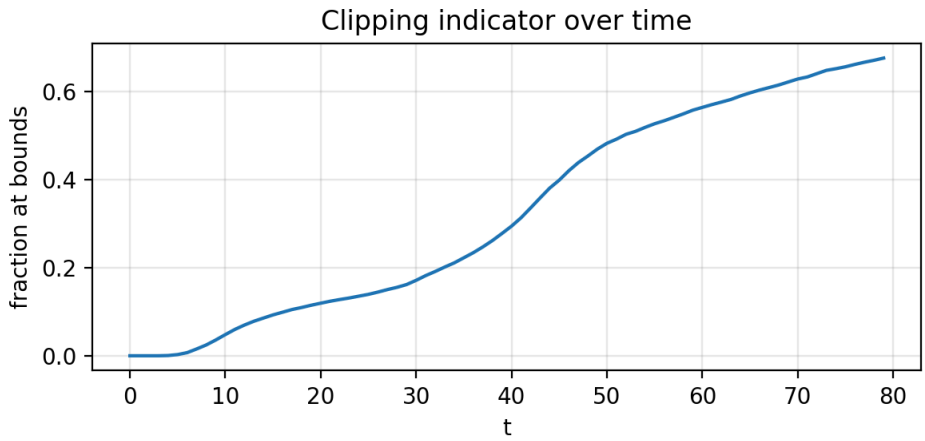

Figure 6. The fraction of agents near the drift bounds over time. A growing fraction near one indicates saturation due to clipping.

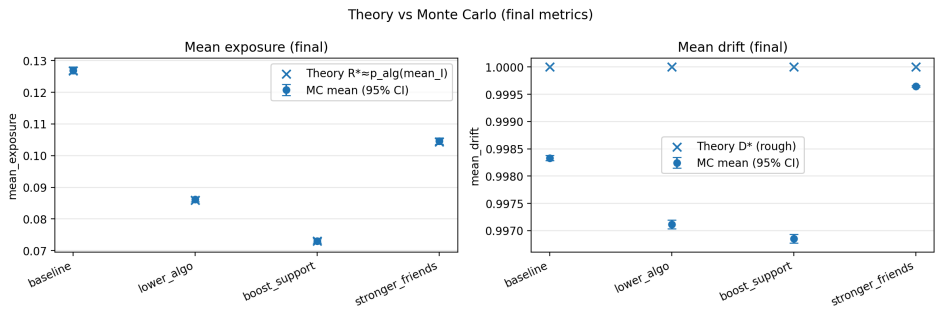

Figure 7. Rough theoretical predictions for final metrics compared to Monte Carlo expected values with confidence intervals. Differences are driven by clipping and by the co-evolution of isolation and social engagement.

Uncertainty analysis

I estimate expected values of each metric by averaging over many Monte Carlo runs. I quantify the remaining Monte Carlo uncertainty using a ninety five percent confidence interval for the expected value. If $x_1, \ldots, x_N$ are the outcomes from N runs, the estimator is the sample mean $\bar{x}$, and the interval is

$$\bar{x} \pm 1.96 \, \frac{s}{\sqrt{N}},$$

where s is the sample standard deviation. When the interval is narrow, the estimated expected value is stable across repeated Monte Carlo experiments. If the interval is wide, more runs are needed. Because the standard error scales like one over the square root of N, reducing the confidence interval half width by a factor of two requires about four times as many simulation runs. In my case, the CIs were quite tight at 200 runs, as you can see from the plot above.

Extension using a stochastic block model network

Erdos Renyi graphs are convenient for theory, but real social networks have clustering. To explore how community structure changes outcomes, I run the same interventions on a stochastic block model that mimics a high school. The network has four communities of two hundred nodes each, with higher within community connection probability than between community probability.

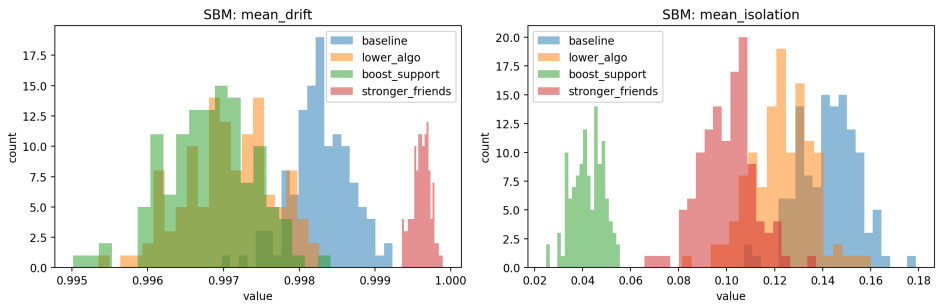

Figure 8. Monte Carlo outcome distributions on the stochastic block model network. The distributions look similar to the ER case.

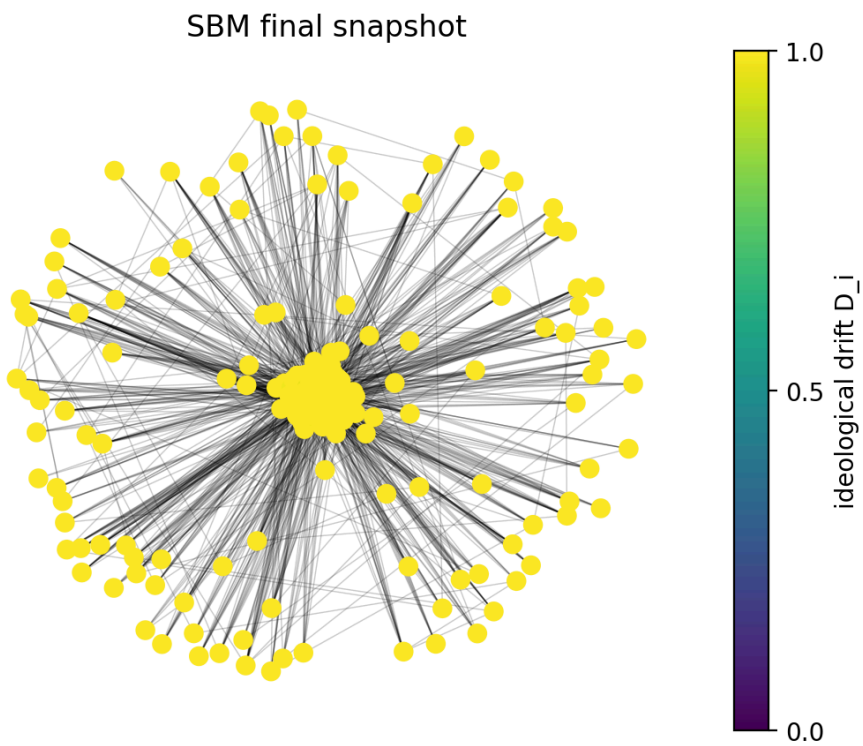

Figure 9. A final network snapshot on the stochastic block model network. Node color represents drift, and the color bar indicates the scale from zero to one.

The qualitative ranking of interventions is similar to the Erdos Renyi case in this experiment. This suggests that the main mechanism is robust to adding clustering, although detailed dynamics can still depend on parameters and on how communities interact.

Recommendations

For school administrators and youth program coordinators, I recommend prioritizing interventions that increase real social support and strengthen social engagement. In this model, increasing δ produces the largest changes. The parameter δ controls how strongly social engagement reduces isolation. Social engagement itself depends on friendship strength and on current isolation, so this result does not mean that support appears automatically. It means that when supportive relationships exist, the ability to stay engaged and to accept help can interrupt the spiral.

A strong program should have two parts. The first part should help people form sustained supportive ties through repeated small group interaction, mentorship, and community spaces that create stable friendships rather than one time events. The second part should reduce the barriers that prevent isolated people from using support. This can include proactive outreach, accessible counseling, and stigma reduction so that accepting help feels safe and normal.

For young men, the same nuance applies. The model does not say that you can simply decide to be less isolated. It says that when supportive relationships exist, staying socially engaged makes it easier to reverse isolation and makes red pill exposure less influential. In practice, this points to small repeated actions that keep relationships alive. You can respond to messages, follow up after social plans, show up regularly to groups where you see the same people, and take low risk steps toward asking for help from friends, mentors, or counselors.

The model also suggests that it is worth actively managing the content environment that shapes exposure. You can reduce red pill exposure by avoiding engagement with rage bait content, by unfollowing accounts that push mysogynistic narratives, and by using settings that reduce recommendation loops. You can also substitute content that supports skill building and real world social life, since algorithms tend to amplify what you repeatedly watch. These choices do not replace real relationships, but they can reduce the pressure that the exposure feedback loop puts on drift while you rebuild social engagement.

In parallel, algorithm changes can reduce the amplification that highly isolated users receive. In the model this corresponds to reducing $p_{\max}$ so that isolation translates into a smaller increase in the probability of being recommended red pill content. The results suggest that combining support building with algorithmic mitigation can be useful. A caveat is that final drift saturates under the current clipping and time horizon. A useful next step would be to measure outcomes before the final timestep, since drift often saturates near one by the end of the simulation, and then choose

parameter values that better match real data on social engagement and red pill exposure.

AI usage: I talked to ChatGPT about my model and asked it to incorporate my ideas into update rules. https://chatgpt.com/share/6921d6dd-ec8c-8010-b654-a0563d3fef61

I used Codex to help me debug, plot, and write mathematical expressions in Markdown cells when creating the report. I asked it if my notebook had any redundancies at the end. It gave me the idea for the extension section:

I asked Codex what it thought about the initial network structure.

My prompt: “How can we make it more realistic? If this models a high school, then ~200 nodes is fine. But then the node topology would probably be more scale-free or with more clustering. If we're modeling a small town, then we'd need ~10000 at least. Then we can get away with more random topology with some clustering. What do you think?”

This is what it said:

“If you want “more realistic,” the biggest win is to pick a setting and choose a network model that matches how friendships actually form (communities + clustering + some hubs), while keeping n small enough that you can still run lots of Monte Carlo trials.

My take

For this assignment, a “high school” (≈200–800 nodes) is the sweet spot: you can justify strong clustering + communities, and you can afford many Monte Carlo runs without optimizing the code heavily. A “small town” at n≈10,000 is plausible conceptually, but with the current implementation (edge updates in Python loops) it will get slow fast, and you’ll end up reducing number of runs so much that your uncertainty analysis gets weaker. More realistic topology options

Community + clustering (high school): use a stochastic block model (SBM): grades/classes/teams as blocks, high within-block connection probability, lower between-block probability. This is usually more realistic than pure scale-free for a school. Hubs + clustering: if you want “popular kids” and triadic closure, use Holme–Kim (scale-free + triangle formation). BA alone is too low-clustering. Spatial realism (small town): random geometric graph or “SBM + spatial mixing” (neighbors more likely to connect) gives local clustering plus some long-range edges. Why this matters for your results

Higher clustering/community structure tends to make drift more local (cluster-level divergence) and makes “bridge” ties disproportionately important. Scale-free-ish hubs make the system more sensitive to a few highly connected nodes (hubs can either stabilize or accelerate drift depending on their D). Recommendation

Frame the report as a high school / small campus sized population and switch the generator to SBM (or Holme–Kim), then add a short robustness check with a larger n (e.g., 1k–2k) if runtime allows. If you really want the “small town” story, we should refactor the edge update to be vectorized/sparse; otherwise it’ll be painful at n=10k.”

From this, I used the SBM idea.